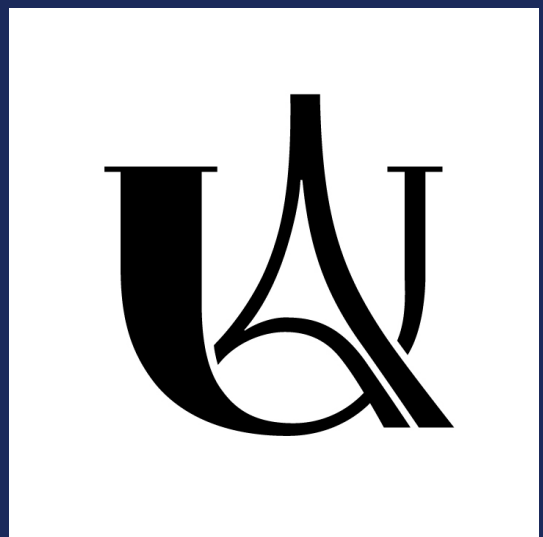




Large Language Models Enable Few-Shot Clustering

A CRITICAL ANALYSIS & ENHANCEMENT PROPOSALS

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Supervised by Pr. Mohamed Nadif & Dr. Imed Keraghel — Université Paris Cité — Based on Viswanathan et al. (CMU, NEC Labs), arXiv:2307.00524

1. Motivation & Problem

Clustering is fundamentally underspecified. Given data $\mathcal{X} = \{x_1, \dots, x_n\}$, there is no unique partition: the same corpus can be grouped by *topic*, *intent*, *sentiment*, or at varying granularity.

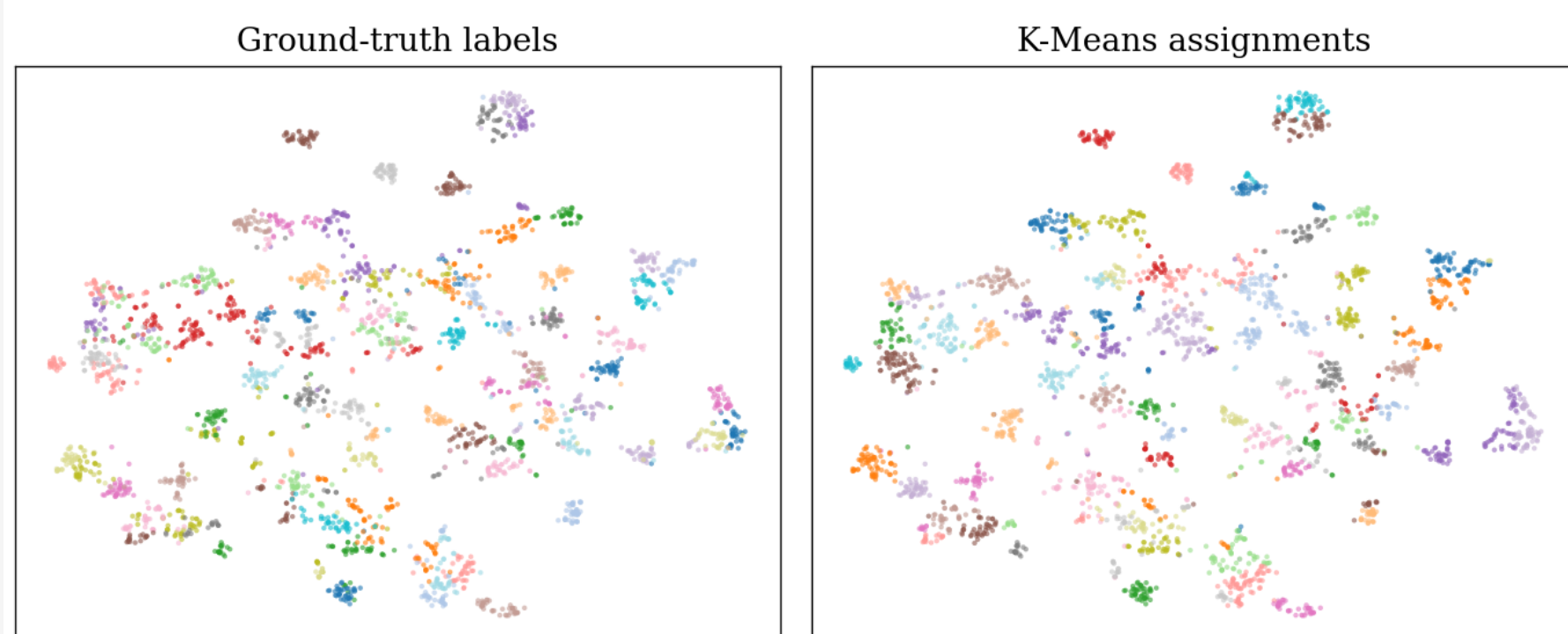
Traditional approach:
20–100 human interactions
(Coden et al., 2017)

LLM approach:
4 demos $\xrightarrow{\text{LLM}}$ 20K constraints
~\$2–5 total cost

Key insight: Can an LLM *amplify* a handful of expert examples into thousands of pseudo-labels to steer clustering?

Baseline: K-Means on Bank77

t-SNE of Bank77 Embeddings (all-mpnet-base-v2)



t-SNE of Bank77 embeddings (all-mpnet-base-v2).

Ground-truth labels (left) vs. K-Means assignments (right).

2. Three-Stage Framework

The paper integrates LLMs at three stages of the clustering pipeline:

BEFORE: Keyphrase Expansion
Enrich representations
1 LLM call / document

↑ **Best**

DURING: Pairwise Constraints
Guide clustering (PCKMeans)
20K pairwise queries

≈ **Good**

AFTER: Post-hoc Correction
Fix low-confidence points
500 LLM queries

↓ **Limited**

Post-Correction: Why It Fails

The LLM **rarely disagrees** with K-Means. When it does, it is often **wrong**: low-confidence $\neq$ misclassified. The LLM is most inaccurate exactly where correction is needed most.

3. Novel Methods: Normalization & Paraphrase Ensemble

LLM Semantic Normalization

Rewrite documents to canonical form *before* embedding:

“Wanna check ma bal”
↓ LLM
“Check account balance”

$$\mathbf{v}'_i = [\mathbf{v}_{\text{orig}} \parallel \mathbf{v}_{\text{normalized}}]$$

✓ Removes slang & abbreviations

✓ **Best ACC on Bank77**

(0.649)

✗ No gain on Tweet (topical data)

LLM Paraphrase Ensemble

Average embeddings over multiple surface variants:

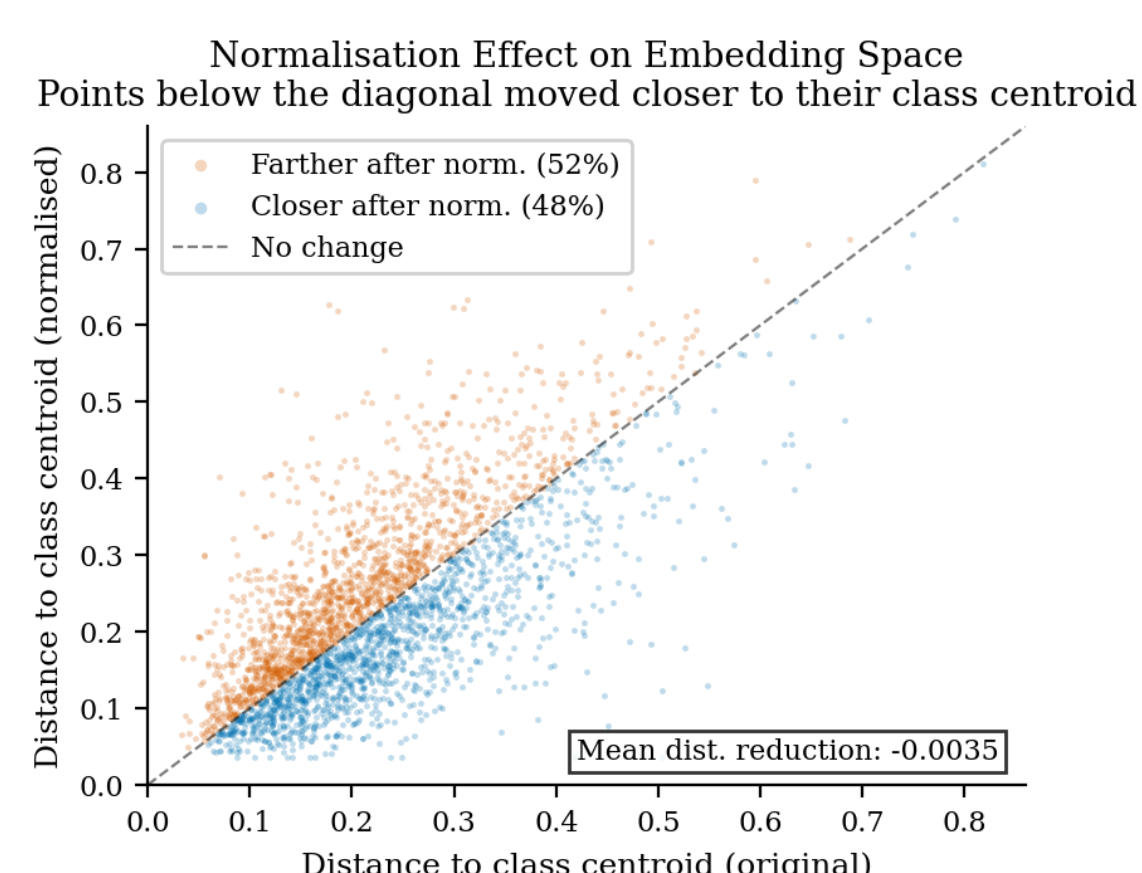
$$\mathbf{v}'_i = \frac{1}{4} \left(\mathbf{v}_i + \sum_{j=1}^3 \mathbf{v}_i^{(j)} \right)$$

Generate 3 paraphrases per doc, embed all, and average to reduce surface noise.

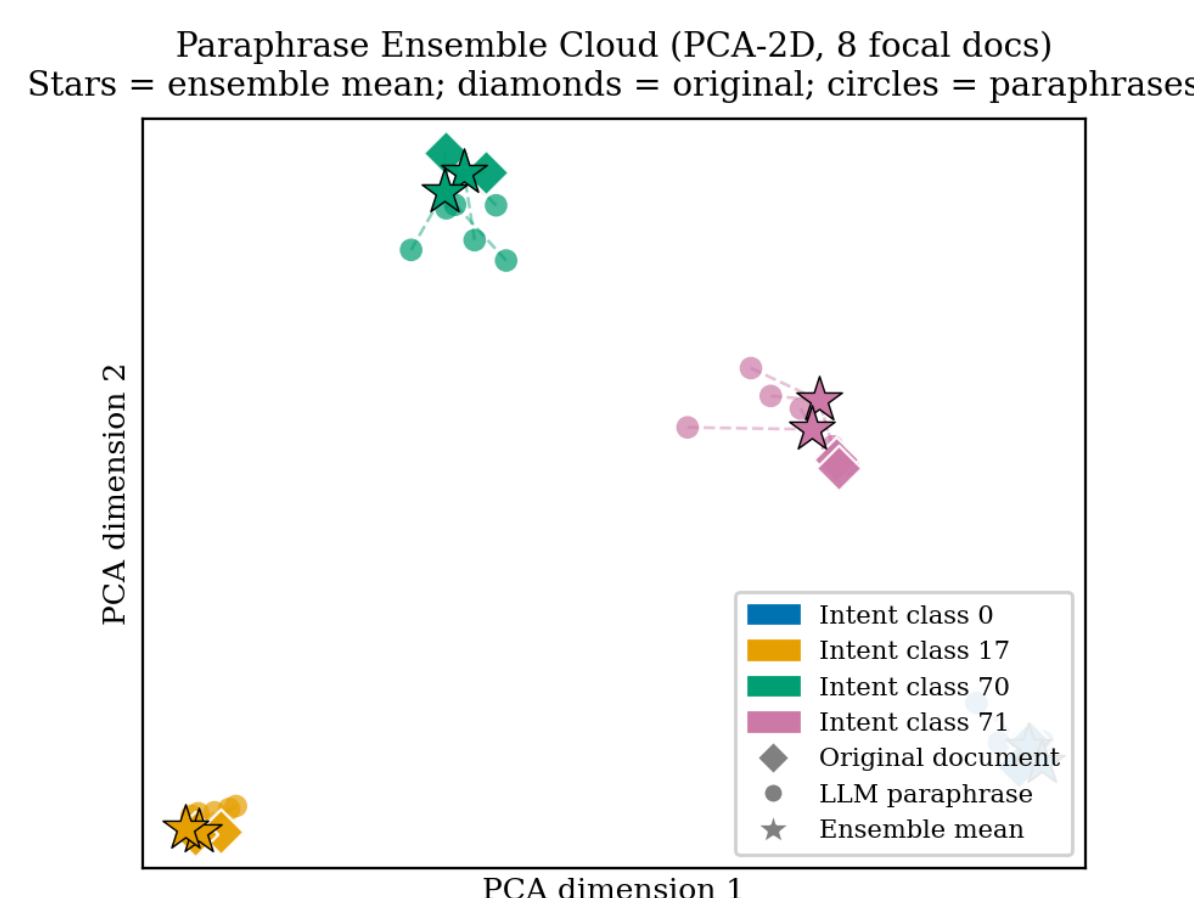
✓ **Best ACC on CLINC** (0.787)

✓ **Best NMI on Bank77** (0.819)

✗ 4× embedding cost



Normalization effect: 48% of points move closer to their class centroid (blue), 52% farther (orange).

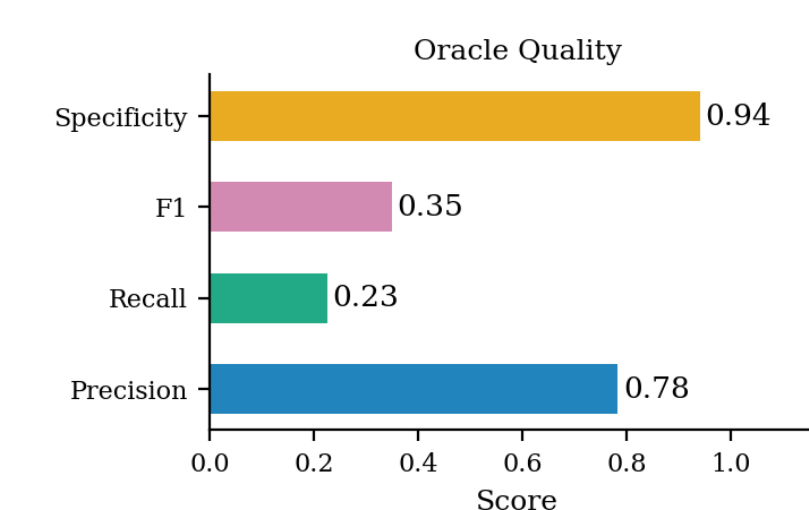


Paraphrase cloud (PCA-2D): ensemble mean (★) stays close to the original (◊).

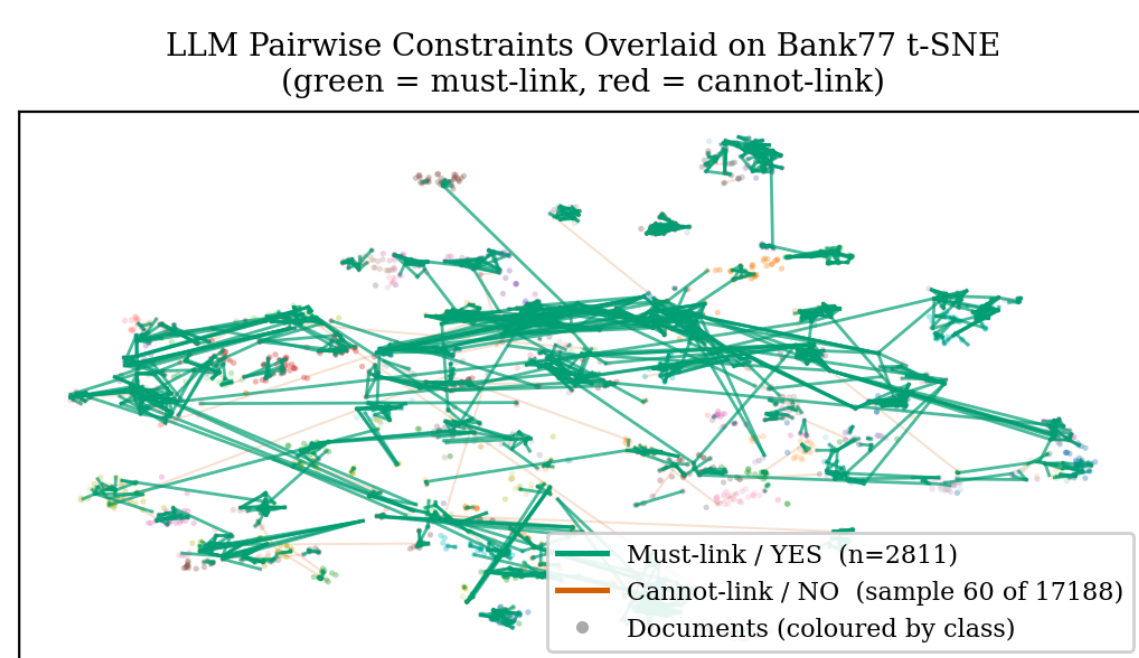
4. LLM Oracle Quality

LLM Oracle Accuracy on Bank77 Pairwise Queries (n = 19,999)

Confusion Matrix (ground-truth columns)	
LLM: YES (must-link)	TP 2202 FP 609
LLM: NO (cannot-link)	FN 7582 TN 9606
Same class Diff. class	

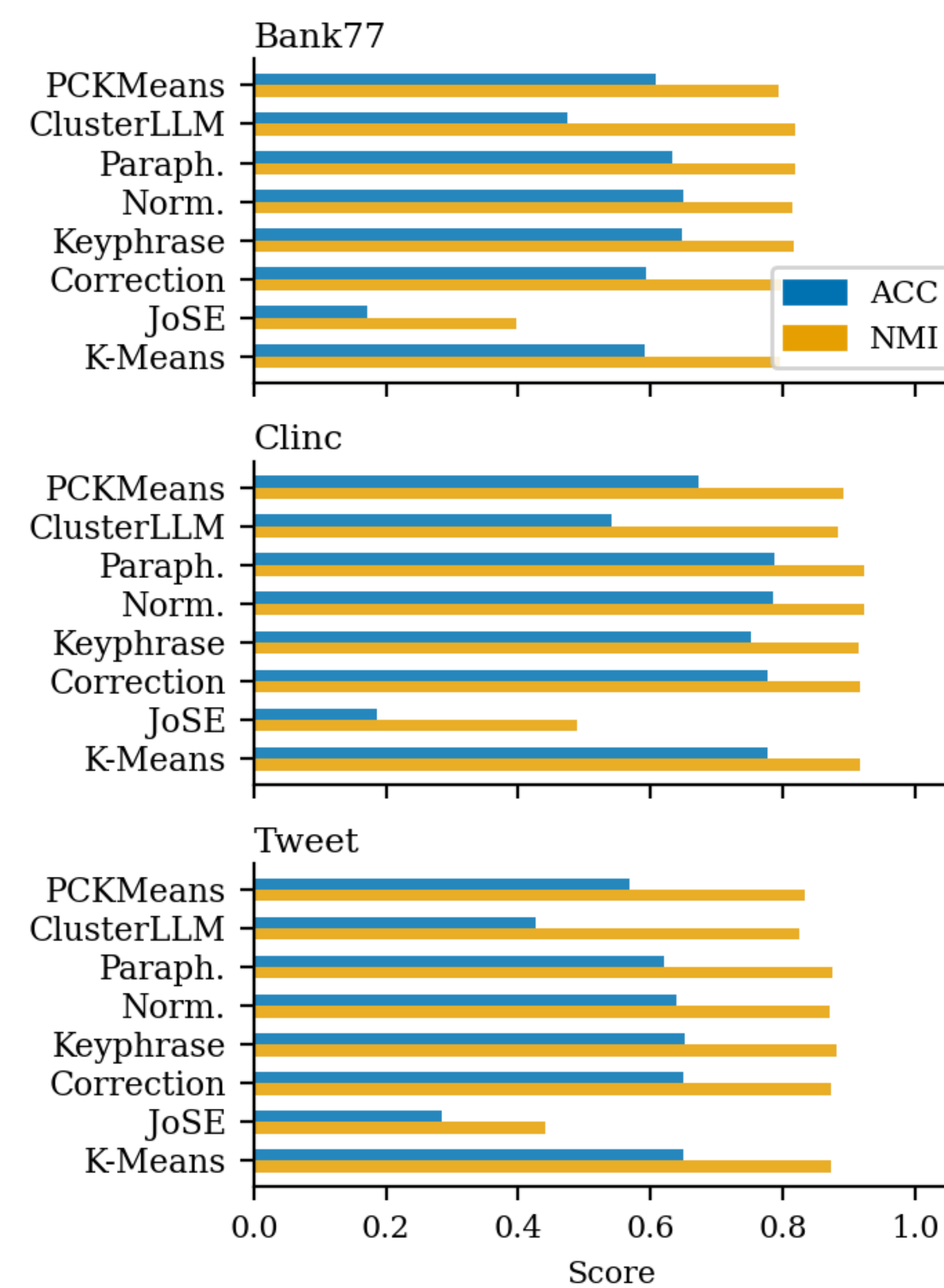


LLM oracle: high precision (0.78) but very low recall (0.23) on Bank77.

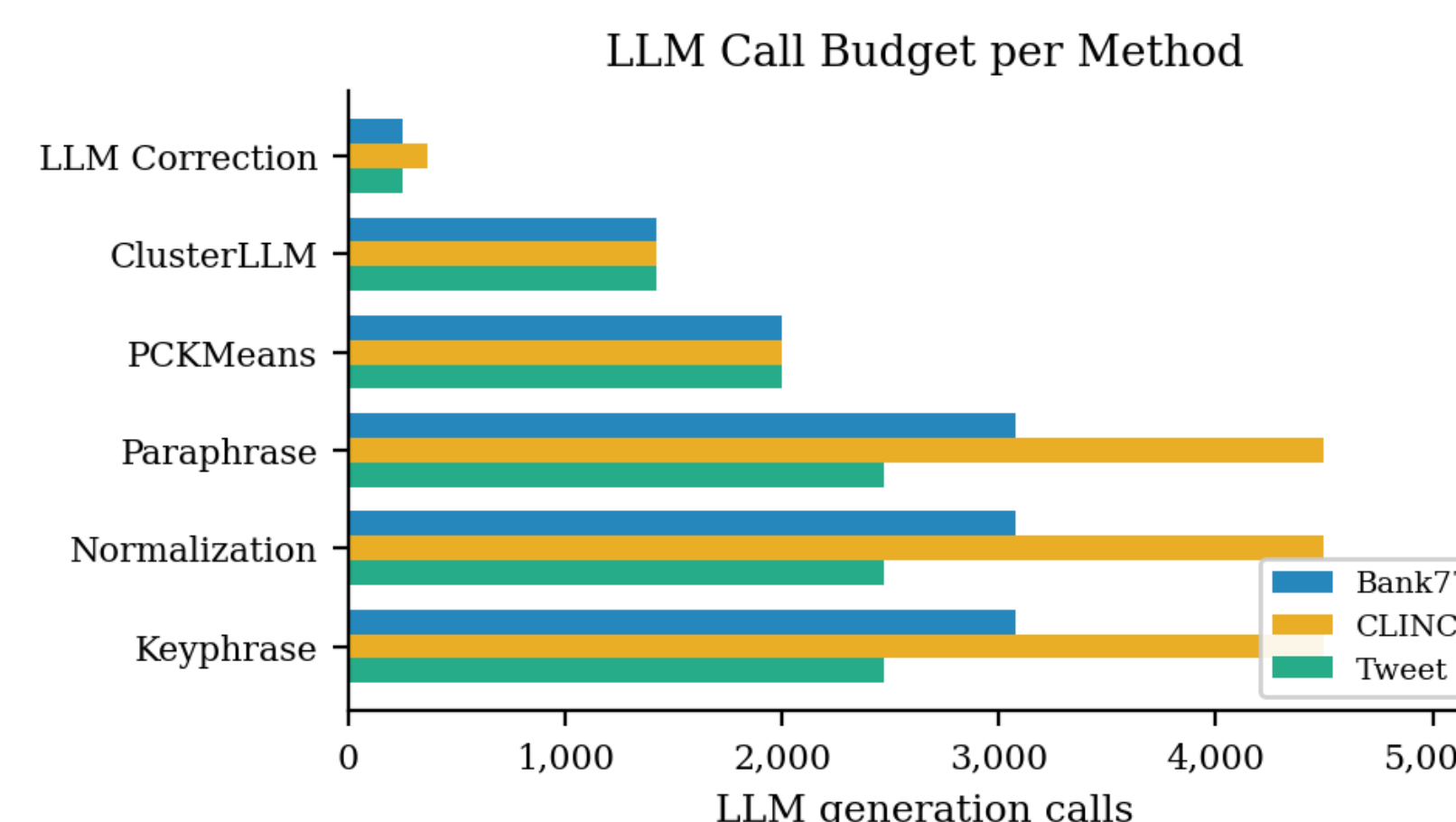


Must-link (green) / cannot-link (orange) constraints overlaid on t-SNE.

5. Full Results



ACC and NMI across all methods on Bank77, CLINC, and Tweet.

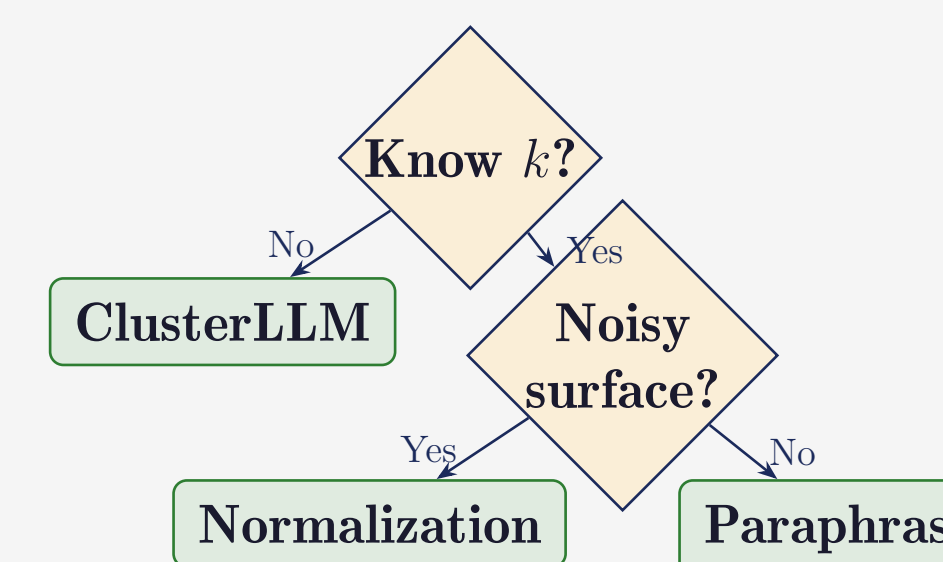


LLM call budget per method: correction is cheapest, keyphrase and paraphrase are moderate.

6. Summary & Recommendations

- 1. Keyphrase expansion works:** simple, cheap, consistent gains across all datasets.
- 2. Novel methods are best:** Normalization & Paraphrase beat all baselines on 2/3 datasets.
- 3. Post-correction is limited:** the LLM is most inaccurate on the uncertain points.
- 4. Per-document is safest:** no constraint propagation errors.

Method Selection Guide



Bottom line: Simple per-document LLM tricks beat complex methods.